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A Longitudinal Study of the Effects of Boundary Spanning Supervision on Turnover and Promotion in Research and Development

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An investigation of the influence of boundary spanning project supervisors on the turnover and promotion of engineering professionals found that, in general, boundary spanning supervisors had little direct effect on the careers of all project subordinates. However, project supervisors who also were technical gatekeepers significantly affected the organizational careers of young engineers.

What factors influence turnover and promotion in research, development, and engineering (RD&E) settings? One research stream has looked at the effects of job experiences and formal supervision on turnover, performance, and promotion (Andrews & Farris, 1967; Berlew & Hall, 1966); others have looked at the norms and climate created in the laboratory (Barth & Vertinsky, 1975; Pelz & Andrews, 1966); and still others have looked at the effects of informal communication networks and boundary spanning roles (Allen, 1977; Katz & Tushman, 1981). In RD&E settings, boundary spanning individuals are key actors in the laboratory's communication and information processing activities (Tushman & Nadler, 1980). The present study focuses on formal and informal aspects of leadership by investigating (over a 5-year period) the influence of boundary spanning supervisors on the turnover and promotions of their project subordinates. Underlying the study is the basic idea that career decisions are strongly affected by how well individuals are linked into their organization's formal and informal networks (Graen & Ginsburgh, 1977; Organ & Greene, 1972).

Literature Review and Hypotheses

To keep informed about relevant external developments as well as new work requirements, project teams must gather information from a variety of outside sources. Such contact is especially critical for RD&E project teams, given their dependence on new technological advances within the larger professional community as well as their need to coordinate with other organizational areas (Roberts & Fusfeld, 1982; Utterback, 1974).

Although RD&E groups must acquire outside information, considerable research has shown that widespread direct contact by all engineering project members is *not* an effective method for communicating outside the project group. Instead, boundary spanning project members are needed to link project teams effectively to outside sources of relevant information. As discussed by Allen (1977) and Katz and Tushman (1979), special project members, labelled gatekeepers, are needed to link project colleagues to key sources of information both inside and outside the organization. Other project members, labelled internal liaisons, link project colleagues only to sources of information within the organization. Outside information is transmitted effectively into project groups through both types of boundary spanning individuals (Tushman, 1977). First, gatekeepers and internal liaisons gather and understand relevant outside information; subsequently they channel it in more meaningful terms to their project colleagues. Boundary spanning individuals, then, perform an extremely valuable function, for they are the principal means by which outside ideas and information are transferred effectively into RD&E project groups (Allen & Cohen, 1969; Katz & Tushman, 1981). Although gatekeepers and internal liaisons connect project colleagues to important information sources within the organization, only gatekeepers provide an effective interface with technical knowledge and advancements outside the organization (Tushman & Scanlan, 1981).

Most boundary spanning individuals in RD&E are also project supervisors, although supervisory status *per se* does not automatically result in high communication activity either within or across organizational boundaries. Rather, project supervisors who are seen as technically competent, who keep up-to-date, and who are seen as valuable sources of ideas and information are likely to become boundary spanning individuals simply because they will be consulted and listened to more frequently on work related matters (Rosen, Billings, & Turney, 1976). Previous research, in fact, has shown that between 70 and 80 percent of the boundary spanning individuals also are first level project supervisors. In contrast, only about half of the project supervisors perform a boundary spanning function (Allen, 1977; Schwartz & Jacobson, 1977; Tushman & Scanlan, 1981).

Even though boundary spanning has been recognized as one of the more important elements of effective leadership and managerial behavior (Graen & Ginsburgh, 1977; Likert, 1967; Mintzberg, 1973), very little is known about how boundary spanning activity relates to other important managerial functions. In addition to enhancing project effectiveness, would boundary

spanning project supervisors also be more likely to affect the work activities and careers of those project members reporting to them?

Research over the past 15 years has demonstrated that interpersonal communications are the primary means by which engineering professionals acquire and disseminate important ideas and information (Allen, 1977; Katz, 1982a; Menzel, 1966). Moreover, it is through such social processes that most professionals come to learn the norms, values, and operating rules of their organization (Katz, 1980; Kerr, 1977). Rather than relying on written technical reports, publications, or other formal structures, professionals in RD&E keep abreast of new technical developments and organizational demands through informal contact and personal associations with other RD&E professionals (Allen, 1977; Farris, 1972).

If interpersonal contact and interaction are key determinants of integration and socialization in RD&E and if boundary spanning supervisors are critical players in the laboratory's communication and information processing network, then these supervisors are doing much more than simply channeling outside information into their project groups. Most likely, they are assuming a broader role within their work groups, becoming actively involved in the training, integration, and socialization of their subordinates. Katz and Tushman (1981) found, for example, that gatekeeping supervisors not only gathered and disseminated external information, but they also improved the external interface of their engineering subordinates by helping to direct, coach, and interpret their external communications.

As a result of their elaborate outside contact and close working relationships with project subordinates, boundary spanning supervisors not only improve their group's technical performance (Tushman & Katz, 1980), but they also may directly affect the personal growth and development of project members. To the extent that boundary spanning supervisors help project members participate and contribute more effectively within their work settings, have clearer working relationships with other corporate areas, and communicate more easily with outside professionals, project members are less likely to leave the organization (Farris, 1971; Graen & Ginsburgh, 1977; Pelz & Andrews, 1966).

Boundary spanning supervisors also are highly influential in technical, administrative, and personnel decision making (Tushman & Romanelli, 1983). Consequently, engineers assigned to work with these supervisors have better chances at gaining increased exposure and more extensive work opportunities. If boundary spanning supervisors are influential and provide their subordinates with better opportunities, then engineers working for these particular supervisors should have a greater rate of promotion. Based on these arguments, the following are hypothesized:

1. *Project members working for boundary spanning supervisors are more likely to remain with the organization than are project members working for supervisors who are not boundary spanners.*
2. *Project members working for boundary spanning supervisors are more likely to receive promotions to management than are project members working for supervisors who are not boundary spanners.*

These turnover and promotion effects are proposed for project members in general. Young employees, however, are more likely to benefit from the socialization and developmental role played by boundary spanning supervisors (Berlew & Hall, 1966). Most turnover occurs within the first few years of organizational employment (Schein, 1978). In addition, engineers usually expect promotions to managerial rank sometime between the ages of 30 and 40 (Dalton, Thompson, & Price, 1982; Ritti, 1971). It therefore is expected that the proposed relationships will be particularly strong for younger project members. Accordingly, the hypotheses will be tested for the full sample of professionals and for project members of different age groups.

Project Task Characteristics

Although the importance of boundary spanning has been well demonstrated, recent research indicates that not all RD&E projects are alike in the way they function or in the way they should be managed (Allen, Tushman, & Lee, 1979). Because of differences in work requirements, there are substantial information processing differences among groups engaged in research, development, and technical service kinds of activities.

In research activity, project members are less constrained by local organizational circumstances because the nature of their work is more universally defined. As a result, research project members can communicate effectively with professional colleagues outside the organization without the help of gatekeepers. In fact, Tushman and Katz (1980) found that research teams relying on boundary spanning gatekeepers were significantly less effective than research teams in which members kept themselves up-to-date and maintained their own external contacts. Nevertheless, research teams must be linked to other organizational areas; and evidence suggests that given the researcher's more cosmopolitan perspective, boundary spanning supervisors are an effective way to ensure their organizational coordination (Allen et al., 1979).

Development projects, on the other hand, involve less universally defined tasks in that their problems and solution strategies are defined and operationalized in terms of the organization's particular strengths, interests, language schemes, and cultural norms. Because of this local orientation, communication across organizational boundaries becomes extremely difficult and ineffective for most development project members (Allen, 1977; Baker, Siegmann, & Rubenstein, 1967). Despite this difficulty, development groups must stay well informed about technical developments outside their organization and must maintain close ties with other organizational areas, marketing and operations in particular. Boundary spanning project members are needed for effective linkage of development groups with these critical information areas (Katz & Tushman, 1981).

Although technical service projects are more local in nature, their work deals with more mature technologies and existing knowledge and products

(Rosenbloom & Wolek, 1970). As a result, members of technical service projects can rely more easily on established practices and procedures. In technical service work, therefore, boundary spanning individuals are not necessary because project members can be kept sufficiently informed about external events and information through formal hierarchies and operating procedures (Tushman & Katz, 1980; Walsh & Baker, 1972).

If boundary spanning supervisors are more critical and influential in development work than in either research or technical service, then their influence on the career outcomes of engineering subordinates also should be particularly important in development projects. As a result, the following hypothesis is suggested:

3. Project work characteristics will moderate the effects that boundary spanning supervisors have on the turnover and promotion of engineering subordinates. The influence of such supervisors will be stronger in development projects than in either research or technical service.

Alternative Comparisons

To what extent are promotions simply a function of working for supervisors who are themselves promoted to higher managerial positions? Webber (1976) suggested that working for highly promotable supervisors enhances one's own chances of promotion. Boundary spanning supervisors usually are promoted (Allen, 1977; Katz & Tushman, 1981), but they comprise only a subset of supervisors who receive promotions over a given time frame. The present study investigates whether there is a particular benefit in working for boundary spanning supervisors rather than for any project supervisor who gets promoted. Furthermore, it is possible that being assigned to a high performing project group affects the turnover and promotion opportunities of young project members more than do the technical and interpersonal skills of their boundary spanning supervisors. The hypothesized relationships, therefore, also are tested and compared against the effects of working in high performing project teams.

Method

Setting

This study was conducted among all project members working in a large corporate RD&E facility. At the start of the study, the facility's professionals ($N = 325$) were divided into seven separate functional departments, which in turn were subdivided into 61 projects organized around specific, long term types of discipline and product focused problems. Each professional was a member of only one project group.

Five years after the initial data collection period, the titles and positions of those professionals still employed by the organization were obtained.

Although the RD&E facility nearly doubled in size during this interval, longitudinal examinations reported here focus on the career histories of only those professionals employed at the start of the study. In addition, 8 percent of the project members retired during the 5-year period and consequently were excluded from all analytical investigations.

Communications, Gatekeepers, and Internal Liaisons

To measure communication activity, project members reported (on specially provided lists) those individuals with whom they had work-related oral communication on a randomly chosen day each week for 15 weeks, including whom they talked to and how many times they talked to that person during the day. Social and written communications were not reported. During the 15 weeks, the overall response rate was 93 percent. Moreover, 68 percent of all communications within the RD&E facility were reciprocally reported by both parties. As discussed by Katz and Tushman (1979), all communication data were used as reported. Given systematic bias in self-report data (e.g., supervisors overreport communication from subordinates), a 68 percent reciprocity rate over many thousands of communications represents a high level of internal reliability and provides a clear, accurate picture of each project member's communication patterns.

For each project member, six mutually exclusive communication measures were operationalized as follows:

1. *Departmental communication*: The amount of communication with other nonsupervisory engineering colleagues within his or her functional department including project colleagues.
2. *Laboratory communication*: The amount of communication with other engineering colleagues within the remaining six functional departments.
3. *Immediate supervisory communication*: The amount of communication with his or her immediate project supervisor.
4. *Departmental supervisory communication*: The amount of communication with his or her departmental supervisor.
5. *Corporate communication*: The amount of communication with other individuals outside the RD&E facility but within other corporate divisions, primarily marketing and manufacturing.
6. *External professional communication*: The amount of communication with other RD&E professionals outside the parent organization, including professionals within universities, consulting firms, and various professional societies.

For each project engineer, the amount of communication to these two horizontal, two vertical, and two outside sources of information was calculated by summing the number of interactions reported during the 15 weeks (see Katz and Tushman, 1979, for details). Except for the high positive correlation between departmental and immediate supervisory communication ($r = .34$), the six measures were not significantly associated with one another.

Conceptually, gatekeepers are defined as those project members who are very high internal communicators and who also maintain very high external contacts with outside professionals. This study operationalized gatekeepers as those project members whose departmental and external professional communications were both in the top fifth of their respective distributions (Katz & Tushman, 1981; Whitley & Frost, 1973). Internal liaisons, on the other hand, were defined as those project members who were in the top fifth of both their departmental communication distribution and their communications to other functional departments and organizational divisions (Allen & Cohen, 1969; Tushman & Scanlan, 1981). Based on these definitions, 18 percent ($n = 11$) of the project supervisors functioned only as internal liaisons, 13 percent ($n = 8$) functioned only as gatekeeping supervisors, and 11.5 percent ($n = 7$) fulfilled both the gatekeeping and internal liaison roles. The remaining project supervisors ($n = 35$) were not performing a boundary spanning function either as a gatekeeper or as an internal liaison. Finally, in one of these 35 cases, the position of project manager was held temporarily by the departmental supervisor. Engineers from this group, as a result, were not included in any statistical comparisons involving prior reporting relationships.

Project Characteristics

As previously discussed, RD&E projects differ widely with respect to the generation of knowledge versus the utilization of existing knowledge and experience. Based on this dimension, project categories were organized around research, development, and technical service kinds of project activities. Using definitions described in Katz and Tushman (1979), project members indicated how well each category represented the objectives of their project work. As in Pelz and Andrews (1966), project members also indicated the percentage of their project activities that fell into each of the possible project categories. A weighted average of the answers to these two questions was calculated for each project member (Spearman-Brown reliability = .91).

To categorize projects empirically, project member responses were averaged to yield 14 research, 24 development, and 23 technical service project groupings. As discussed by Tushman (1977), the homogeneity of members' perceptions of their project characteristics also was checked to ensure the appropriateness of aggregating across individual scores.

Project Performance

To get comparable measures of project performance across significantly different technologies, all departmental and laboratory managers ($N = 9$) were interviewed individually and asked to evaluate the overall technical performance of all projects with which they were sufficiently familiar. When they could not make an informed judgment, they did not rate that project.

Each project was independently evaluated by an average of five managers using a 7-point Likert type scale ranging from (1) very low to (7) very high. Individual ratings were averaged to yield overall project performance scores (Spearman-Brown reliability = .81). To classify project members according to whether they were working in a high or low performing project team, project groups were split at the sample mean of 4.59.

Promotion and Turnover

Almost five years after the collection of the preceding data, data on managerial promotions and turnover were gathered. Despite the facility's strong growth, 31 percent of the project members and 19 percent of the project supervisors had left the company during this time interval. Furthermore, among the 15 gatekeepers and 18 internal liaison supervisors, the turnover rates were 20 and 17 percent, respectively.

In this organization, managerial positions and titles start within the department above the project supervisory level. During the 5-year interim period, 11 percent of the project members and 46.5 percent of the project supervisors had been promoted to management positions. Although less than half of the project supervisors had received management promotions, 73.3 percent of the gatekeeping subset and 67 percent of the internal liaison subset had been promoted to management levels.

Results

Turnover

The first hypothesis was concerned with the influence of boundary spanning supervisors on the turnover rates of project engineers. Table 1 reports the percentages of project members who remained with the organization over the 5-year period as a function of their prior type of supervision. For the sample as a whole, project members who reported to gatekeeping supervisors had a significantly lower rate of organizational turnover than did engineers assigned to supervisors who were either nonboundary spanners or only internal liaisons. Engineers who reported to internal liaisons had the lowest retention rate over the 5-year period (only 62 percent); those who reported to gatekeeping supervisors or supervisors who were *both* gatekeepers and internal liaisons had the highest retention rates (82 and 85 percent, respectively). Clearly, gatekeepers had a more positive effect on retention rates than did internal liaisons. Furthermore, project members assigned to those supervisors who received promotions did not stay with the organization any longer than did members who worked for unpromoted supervisors (Table 1B). Similarly, engineers who worked in high performing projects did not remain with the organization any longer than did those who worked in low performing projects (Table 1C).

Table 1
Supervisory Influence on the Proportion of Engineers Remaining
in the Organization Over a 5-Year Period

<i>A. Prior Type of Reporting Relationship: Engineer Worked for a Supervisor Who Was:</i>				
	<i>Internal Liaison</i>	<i>Gatekeeper</i>	<i>Gatekeeper and Internal Liaison</i>	<i>Neither a Liaison nor a Gatekeeper</i>
% Remaining in organization after 5 years (N)	62% ^b (45)	82% ^a (28)	85% ^a (20)	70% ^b (115)
<i>B. Did Engineer Work for a Supervisor Who Was Subsequently Promoted to a Managerial Position?</i>		<i>C. Did Engineer Work in a Project That Was:</i>		
	<i>Yes</i>	<i>No</i>	<i>High Performing</i>	<i>Low Performing</i>
% Remaining in organization after 5 years (N)	67.6% (102)	71.7% (106)	73.7% (114)	63.5% (104)
	Not significantly different		Not significantly different	

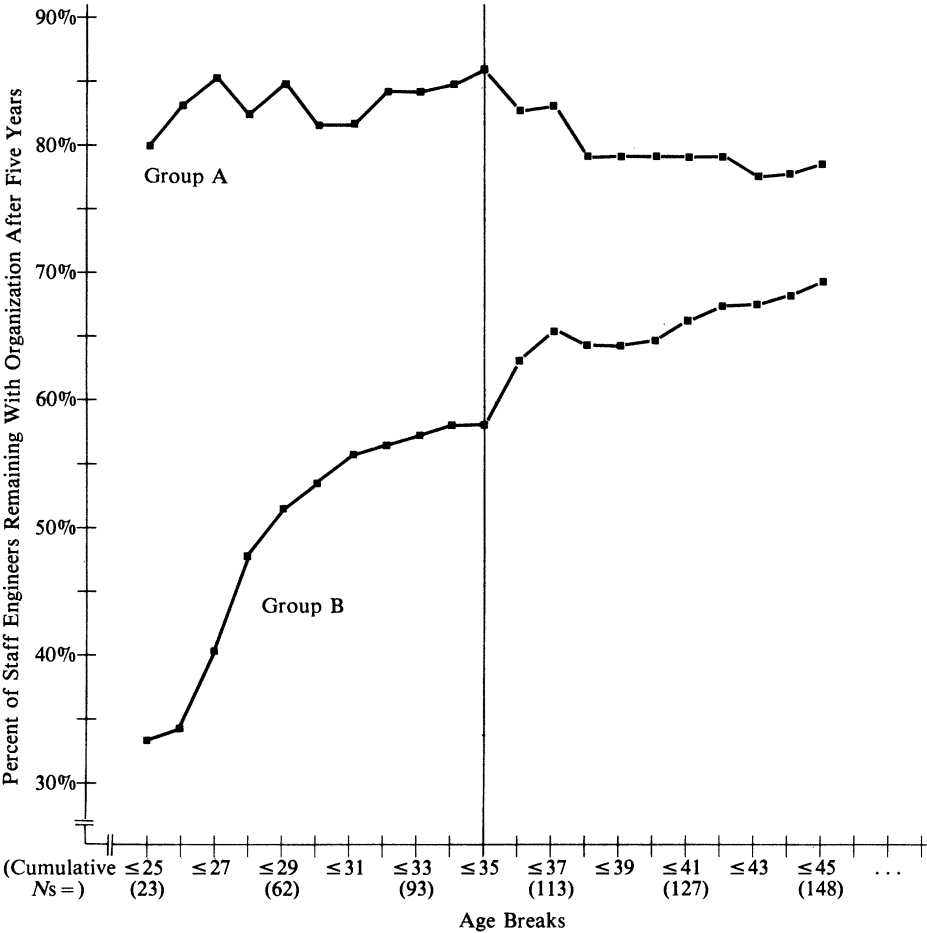
^{a,b}Based on *t*-tests, percentages superscripted "a" are significantly greater than those superscripted "b" ($p < .10$).

N: The Ns in sections A and B are based on engineers from only 60 projects. In section C, N is based on engineers from all 61 projects.

Because 70 percent of the turnover occurred for project members less than 36 years of age, additional comparisons were carried out for separate age groupings. Except for gatekeeping supervisors, these additional comparisons failed to uncover any significant turnover differences among prior types of reporting relationships or between high and low project performance. To pinpoint the influence of gatekeepers on subordinate turnover, Figure 1 plots as a function of age the cumulative retention rates of project members reporting to gatekeeping supervisors (Group A) and those members not reporting to gatekeeping supervisors (Group B). In Figure 1, as well as in the rest of this paper, the term gatekeeping supervisors refers to supervisors who were either functioning only as gatekeepers or functioning simultaneously as gatekeepers and internal liaisons. Nongatekeeping supervisors refers to the rest of the supervisory population.

Of those project subordinates who were age 25 or less, only 33 percent remained in the organization if they had *not* reported to a gatekeeping supervisor. The comparable percentage for project members assigned to a gatekeeping supervisor was almost 80 percent. Similarly, of those project subordinates who were 35 years old or less, only 57 percent remained with the organization if they had *not* been working with a gatekeeping supervisor. The comparable percentage in Group A was 84 percent. Although this difference is statistically significant, the figure clearly shows that most of the difference in retention rates between Groups A and B occurs among members between the ages of 25 and 29. Retention rates between the two groups converge rather quickly after age 35.

Figure 1
Retention of Engineers After Five Years
by Prior Type of Reporting Relationships at Successive Age Breaks^a



^aGroup A: staff engineers who reported directly to a gatekeeping supervisor; Group B: staff engineers who reported directly to a nongatekeeping supervisor. The vertical line indicates that pairwise percentages between Groups A and B remained significantly different ($p < .05$) through an age break of 35 years. Ns represent the total number of engineers within both groups at six representative age breaks.

Based on the curves in Figure 1, there appear to be at least three distinct age breaks: (1) less than 30; (2) 30-35; and (3) 36 or more. Using these age breaks, Table 2 shows that project subordinates less than 30 years of age were significantly more likely to remain in the organization if they had been working with a gatekeeping supervisor. Over 84 percent of these engineers were still with the organization after five years versus only 51 percent for the parallel group of young engineers not reporting to gatekeeping supervisors.

Table 2
Proportion of Engineers Remaining in the Organization
After Five Years by Prior Age and Reporting Relationship

<i>Prior Age of Engineers</i>	<i>Prior Reporting Relationship Percent</i>		<i>Percent Differences</i>
	<i>Assigned to a Gatekeeping Supervisor</i>	<i>Not Assigned to a Gatekeeping Supervisor</i>	
Less than 30	84.6 (<i>n</i> = 13)	51.0 (<i>n</i> = 49)	33.6*
Between 30 and 35	83.3 (<i>n</i> = 12)	66.7 (<i>n</i> = 27)	16.6
Greater than 35	78.3 (<i>n</i> = 23)	78.6 (<i>n</i> = 84)	.3

**p* < .05

Similar significant differences in turnover, however, did not emerge between project members for each of the two remaining age categories.

In project groups, individuals functioning as gatekeepers are not always the project's supervisor. In the sample, there were five projects in which one of the members, and not the project supervisor, was functioning as a gatekeeper. Although this number of cases is rather small, the turnover pattern of engineers from these five groups was nearly the same as the turnover pattern from projects without any gatekeeper. For members less than 35 years of age, for example, the 5-year retention rate from projects without gatekeepers was 56 percent; the comparable retention rate from the five projects that had only a nonsupervisory gatekeeper was also low, only 58 percent. The retention rate with a gatekeeping supervisor was significantly higher at 84 percent. Although collegial gatekeepers are important in obtaining needed technical information, apparently only a gatekeeping supervisor has strong positive effects on early socialization and developmental processes.

Although gatekeeping supervisors may have led to reduced turnover levels among young project subordinates, Hypothesis 3 suggests that these effects might be stronger for members of development project groups. This possibility is reported in Table 3, which compares the turnover rates of project engineers less than 30 years of age for each project category. The turnover differences between project subordinates with and without gatekeeping supervisors was consistently positive for each project area, although the small number of cases precludes statistical testing. For each project area, approximately half of the young project members not reporting to a gatekeeping supervisor had left the company within five years despite substantial growth within the RD&E facility. In sharp contrast, none of the corresponding turnover rates for project members reporting to a gatekeeping supervisor exceeded 25 percent.

Clearly gatekeeping supervisors had considerable influence over the turnover rates of young professionals within this facility. What is it about gatekeeping supervisors that brings about these lower levels of turnover? As previously discussed, turnover may be a function of how well young

Table 3
Proportion of Young Engineers (<30) Remaining in the Organization After Five Years by Prior Reporting Relationship and Project Task Areas^a

<i>Prior Project Areas</i>	<i>Prior Reporting Relationship Percent</i>		<i>Percent Differences</i>
	<i>Assigned to a Gatekeeping Supervisor</i>	<i>Not Assigned to a Gatekeeping Supervisor</i>	
Applied research	83.3 (n = 6)	44.4 (n = 9)	38.9
Product/process development	75.0 (n = 4)	53.8 (n = 26)	21.2
Technical service	100.0 (n = 3)	50.0 (n = 14)	50.0

^aBecause of small sample sizes, statistical tests are not used; instead, the focus is on the overall consistent pattern of positive proportional differences.

professionals get integrated into their organization's formal and informal networks. Because gatekeepers are key individuals in these networks, reporting to a gatekeeping supervisor may well facilitate the young professional's linkages to important information sources both within and outside the organization. An investigation was made as to whether young project members reporting to gatekeeping supervisors had different interaction patterns than did project members reporting to nongatekeeping supervisors. Table 4 reports mean communication scores broken down by supervisory relationships and turnover. Separate comparisons are reported for communication to each of the six information sources used in the study.

As shown in Table 4, there were no significant differences between the communication patterns of young engineers who worked for gatekeeping supervisors and those who worked for nongatekeeping supervisors. Their levels of interaction were very similar in each communication area. Furthermore, regardless of their supervisory reporting relationship, project members who either stayed or left did not differ in the intensity of their prior horizontal collegial interactions or in their contacts with individuals and professionals outside the RD&E facility. What differentiated young stayers from leavers was their level of contact with their project and departmental supervisors; that is, their degree of vertical communication and integration. For both gatekeeping and nongatekeeping relationships, project subordinates who remained over the five years had significantly more communication with their departmental supervisor than did project members who left. Stayers also had more interaction with their project supervisors, although the difference was not quite significant for members who reported to nongatekeeping project supervisors.

What are the relative effects of prior communication and reporting relationships on subsequent turnover? A hierarchical discriminant analysis was run on the turnover outcomes of the 62 project members less than 30 years of age using the six communication measures and the prior reporting

Table 4
Mean Communications of Young Engineers (<30 years) Broken Down by Prior Reporting Relationship and Subsequent Turnover

<i>Prior Measures of Communication (per Month)</i>	<i>Prior Reporting Relationship</i>	
	<i>Assigned to a Gatekeeping Supervisor</i>	<i>Not Assigned to a Gatekeeping Supervisor</i>
<i>Horizontal communications with:</i>		
1. Departmental colleagues		
a. Engineers who remained	117.8	104.8
b. Engineers who left	110.2	113.4
2. Laboratory colleagues		
a. Engineers who remained	28.2	26.3
b. Engineers who left	27.1	24.8
<i>Vertical communications with:</i>		
3. Immediate supervisor		
a. Engineers who remained	30.1 ^a	27.6
b. Engineers who left	8.8 ^b	19.0
4. Departmental supervisor		
a. Engineers who remained	4.9 ^a	3.6 ^a
b. Engineers who left	0.0 ^b	1.2 ^b
<i>Outside communications with:</i>		
5. Other corporate areas		
a. Engineers who remained	17.9	20.8
b. Engineers who left	25.8	17.4
6. External professionals		
a. Engineers who remained	1.1	2.1
b. Engineers who left	5.8	2.0

^{a,b}For each pair, communication means superscripted “a” are significantly greater than those superscripted “b” (*p* < .05). Ns can be found in Table 2.

relationship (coded “1” and “0” for working with and without a gatekeeping supervisor) as possible differentiating variables. After both measures of vertical communication entered the discriminant analysis significantly, none of the remaining independent variables, including reporting to a gatekeeping supervisor, accounted for any significant amount of additional discrimination between stayers and leavers. Using the Wilks’ Lambda criterion, the discriminant function containing both vertical communication variables was significant at the *p* < .05 level ($\chi^2 = 6.13$; *df* = 2) and correctly classified 74 percent of the project subordinates. Thus, it is not the assignment of young project members to a gatekeeping supervisor per se that enhances long term retention. What really makes the difference is the high level of vertical interaction that takes place between gatekeeping supervisors and their young engineering subordinates.

In testing the hypotheses, it is important to mention that age really serves as a surrogate means for examining employees at similar career stages. Strictly speaking, several cohorts of employees at different career stages would be the most advantageous method for testing the influence of gatekeeping on project subordinates. In a field study of this sort, however, such cohorts are unavailable. Furthermore, there simply were not enough turnover cases, especially with gatekeeping supervisors, even to begin to determine whether age, tenure, or some other career variable is most appropriate. In any event, age and tenure are so highly congruent for very young employees that any

distinction would be primarily conceptual rather than yielding any clear empirical differences.

Managerial Promotion

During the five years, 23 of the project members were promoted to managerial positions. Table 5 reports promotion percentages for different types of prior reporting relationships. As with turnover, project members who reported to gatekeeping supervisors had a significantly higher rate of promotion than did project members who reported to other kinds of supervisors, including internal liaisons. Furthermore, individuals assigned to supervisors who were subsequently promoted were no more likely to get promoted than were individuals who worked for unpromoted supervisors. Similarly, individuals from high performing projects did not receive proportionately more promotions than did individuals from low performing projects.

Table 5
Supervisory Influence on the Proportion of Engineers Promoted to Managerial Level Positions Over a 5-Year Period

A. Prior Type of Reporting Relationship: Engineer Worked for a Supervisor Who Was:				
	Internal Liaison	Gatekeeper	Gatekeeper and Internal Liaison	Neither a Liaison nor a Gatekeeper
% Promoted to managerial positions after 5 years	11.1% ^b	14.3% ^a	15.0% ^a	11.3% ^b
B. Did Engineer Work for a Supervisor Who Was Subsequently Promoted to a Managerial Position?				
	Yes	No	C. Did Engineer Work in a Project That Was:	
			High Performing	Low Performing
% Promoted to managerial positions after 5 years	11.8%	10.4%	9.6%	12.5%
	Not significantly different		Not significantly different	

^{a,b}Based on *t*-tests, percentages superscripted "a" are significantly greater than those superscripted "b" (*p* < .10).
N: The Ns in sections A and B are based on engineers from only 60 projects. In section C, N is based on engineers from all 61 projects.

Because almost 70 percent of the project members promoted to managerial positions were between the ages of 27 and 32 at the start of our study, the data within this more limited age range were reanalyzed. As before, the various types of reporting relationships revealed no important effects on promotion rates within this restricted subsample, except for the comparison of gatekeeping versus nongatekeeping supervisors (Table 6). For this

Table 6
Proportion of Engineers Promoted to High Level
Managerial Positions Over the Next Five Years by
Prior Reporting Relationship and Project Task Areas^a

<i>Prior Project Areas</i>	<i>Prior Reporting Relationship Percent</i>		<i>Percent Differences</i>
	<i>Assigned to a Gatekeeping Supervisor</i>	<i>Not Assigned to a Gatekeeping Supervisor</i>	
Across all areas	41.2 (<i>n</i> = 17)	17.4 (<i>n</i> = 46)	23.8*
By project area			
Applied research	33.3 (<i>n</i> = 6)	20.0 (<i>n</i> = 10)	13.3
Product/process development	66.7 (<i>n</i> = 6)	18.5 (<i>n</i> = 27)	48.2**
Technical service	20.0 (<i>n</i> = 5)	11.1 (<i>n</i> = 9)	8.9

^aTable includes only those engineers in the age range (27 through 32) in which almost 70 percent of the promotions took place.

**p* < .10

***p* < .05

comparison, the difference was significant; for project members who reported to a gatekeeping supervisor, the promotion rate was 41.2 percent, whereas only 17.4 percent of the engineers who reported to a nongatekeeping supervisor were similarly promoted.

Although proportionately more engineers who had reported to gatekeepers were promoted to management within this general subsample, Table 6 shows that most of this difference can be found in the area of product and process development. No significant advantage was found in the promotion rates of project engineers reporting to gatekeepers in either research or technical service areas. On the other hand, two-thirds of the engineers reporting to gatekeepers in development projects received management promotions, in contrast to only 18.5 percent of the engineers reporting to nongatekeepers, although once again the number of cases is rather small. Nonetheless, development work is precisely the project area in which gatekeepers presumably are most necessary and influential and in which they were hypothesized to have the strongest influence over managerial promotions.

As in the turnover analyses, the communication patterns of project members within the 27-32 year old age range were examined to see if those promoted also had differential patterns of contacts and interactions within their work settings. None of the communication measures, however, was significantly related to managerial promotions for these individuals.

Discussion

The research findings presented here support the idea that supervisory behavior is an important factor in the making of one's organizational and professional career. Not all supervisors in the study, however, had

comparable relationships with the career outcomes of their engineering subordinates. Only boundary spanning gatekeeping supervisors were significantly associated with reduced turnover rates and higher rates of subordinate promotion to management. These associations, moreover, were particularly strong for young professionals. They disappeared for older, more experienced engineers.

On a percentage basis, more young engineers who had gatekeeping supervisors remained with the firm than did those whose supervisors were not gatekeepers. Why were these gatekeepers so strongly related to lower levels of turnover? Was it the result of their supervisory status, their technical ability, or their outside professional contacts? These characteristics certainly are important, but the results on communication suggest that gatekeepers were related to the low turnover among their young project subordinates because of the high levels of hierarchical interaction and activity that took place between them. Because gatekeeping supervisors are technically competent, interpersonally active, and they readily identify with the professional orientations of young engineers, they establish close working relationships with most of their project subordinates, almost 85 percent in the study. It was this high level of hierarchical activity and concern that discriminated between young engineers who stayed and those who left. In the relatively few cases in which gatekeeping supervisors either failed to communicate or denied access to their young subordinates, these individuals probably became disenchanted, gave up, and left. Thus, it may not be the gatekeeping role or supervisory status *per se* that is related to turnover. In line with the findings of Graen and Ginsburgh (1977), what seems most beneficial are high levels of work-related contact and involvement with relevant competent supervisors—interactions that occur most frequently with gatekeeping supervisors.

In addition to these turnover relationships, gatekeeping supervisors were linked to the managerial promotions of project subordinates who were between the ages of 27 and 32 at the start of the study. Within this age range, project members working for gatekeeping supervisors attained a significantly higher rate of promotion to management than did project members working for nongatekeeping supervisors. Furthermore, the promotion rate for young project members reporting to gatekeepers in development areas was more than three times the promotion rate of development project members assigned to nongatekeeping supervisors. In development work, gatekeepers are highly influential individuals who strongly enhance project performance by connecting engineers to more useful ideas and information outside the project. Having better access to critical information, along with working for influential supervisors, may be associated with greater work opportunities and organizational visibility which, in turn, lead to higher rates of management promotion.

Implications and Conclusions

From a broader perspective, the relationship between lower turnover and high interpersonal involvement with gatekeeping supervisors affirms the

important role that project supervisors can and should play during the early socialization years of young professionals. As discussed by Schein (1978) and Katz (1980), young employees build perceptions of their work environment and establish their new organization identities through the plethora of interactions and interpersonal activities that take place during the early years of their laboratory integration. Young engineers, therefore, not only need to interact with their colleagues and peers, but they also require considerable interaction with and feedback from relevant supervisors to learn what is expected of them and to decipher how to be a high performing contributor.

Because they are well-connected professionally and organizationally, gatekeepers are particularly qualified to meet the breaking-in concerns of young professionals, directing and coupling the professional orientations of young engineers with a more appropriate organizational focus. Most likely, the high level of interpersonal contact between gatekeeping supervisors and young project engineers not only facilitates socialization but also results in more accurate expectations, perceptions, and understanding about one's role in the project and in the larger organization—all of which are important in decreasing the turnover of newcomers (Katz, 1980; Pondy, 1978; Wanous, 1980).

These results, then, combined with other research evidence—for example, Dalton et al. (1982), Graen and Ginsburgh (1977), and Vicino and Bass (1978)—indicate that becoming an integral part of an organization's communication and information processing networks and learning the organization's customs and norms are critically important during the early stages of employees' careers. Although socialization can take place in many different ways, gatekeeping supervisors appear particularly important in the organizational integration of young engineering subordinates, helping them understand and interpret the reality of their new settings in order to function more fully and meaningfully within the organization. Gatekeeping supervisors, then, fulfill an important leadership and training function for young engineers, operating effectively as important socializing agents and network builders for these young employees.

Organizations have to recognize that the problems and concerns of young engineers are real and must be dealt with before they become effective organizational members. Although specific training programs could be developed to teach project managers how to "break-in" the young engineer more effectively, the careful selection of supervisors for young professionals also would go a long way toward alleviating many of the problems that usually occur during this joining-up process.

It also is important to note that gatekeeping supervisors were not related to the career outcomes of their older employees. Individuals in the innovation or stabilization phases of their careers have a more clearly developed sense of personal identity and organizational reality—their reputations and areas of contribution being more firmly defined (Katz, 1982b). As a result, the turnover and promotion rates of these more established veterans may

be influenced more by individual differences and by task and organizational factors than by the characteristics of their immediate supervisors.

Finally, one should realize that in a longitudinal field study of this sort, the random assignment of project members to gatekeeping and nongatekeeping supervisors was not possible. Although the thinking here emphasizes the direct role that gatekeepers might play in influencing project members' careers, it also is possible that gatekeepers either attracted or were assigned members who were more likely to stay or were of higher promotion potential. Furthermore, other uncontrolled organizational factors could have influenced the results. For example, data were not collected on how long project members worked for their respective supervisors. What is known is that project members who reported to gatekeeping supervisors early in their careers had more successful organizational outcomes. It remains for future research to look even more closely at these types of relationships. Even with these traditional caveats, substantial research and practice strongly indicate that gatekeepers are extremely important in RD&E settings, not only for the effective transfer and processing of outside technical information, but also in the socialization and development of young engineers.

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